1. The weekly revenue for a product is given by $R(x) = 145.2x - 0.0165x^2$, and the weekly cost is $C(x) = 8000 + 72.6x - 0.033x^2 + 0.00001x^3$, where x is the number of units produced and sold.

(a) How many units will give the maximum profit?
(b) What is the maximum possible profit?

(a) The number of units that will give the maximum profit is ______.
(Round to the nearest whole number as needed.)

(b) The maximum possible profit is $______.

2. The size of the civilian labor force (in millions) for selected years from 1950 and projected to 2050 can be modeled by $y = -0.0001x^3 + 0.0058x^2 + 1.43x + 56.7$, where x is the number of years after 1950.

(a) What does this model project the civilian labor force to be in 2015?
(b) In what year does the model project the civilian labor force to be 155 million?

(a) In 2015 the model projects the civilian labor force to be ______ million.
(Type an integer or decimal rounded to one decimal place as needed.)

(b) The model projects the civilian labor force to be 155 million in ______
(Round up to the next year.)
.

3. The adjacent table gives the number of worldwide Internet users, in millions.

Complete parts **(a)** through **(b)**.

Full data set

Year	Users (millions)	Year	Users (millions)
1995	17	2002	597
1996	34	2003	721
1997	74	2004	819
1998	151	2005	1025
1999	245	2006	1101
2000	367	2007	1229
2001	539		

(a) Find the cubic function that is the best fit for the data, with y equal to the number of millions of users and x equal to the number of years from 1990.
y = ______
(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

(b) Use the model to predict the number of users in 2016.
The number of users in 2016 will be approximately ______ million.
(Round to the nearest whole number as needed.)

4. Solve the system of equations by any convenient method, if a solution exists.

$$\begin{cases} 4x + 4y = 2 \\ x = 2y + 4 \end{cases}$$

Select the correct answer below and, if necessary, fill in any answer boxes to complete your choice.

- **A.** The solution is x = ______ and y = ______. (Type integers or simplified fractions.)
- **B.** There are infinitely many solutions.
- **C.** There is no solution.

*5. Solve the following system.

$$\begin{cases} \frac{5}{4}x - \frac{4}{3}y + \frac{5}{4}z = 7 \\ \frac{5}{16}x + \frac{3}{4}y - \frac{3}{16}z = -7 \\ \frac{3}{4}x - y + \frac{3}{4}z = 6 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- **A.** The system has a single solution. The solution set is $\{(\square,\square,\square)\}$.
  (Simplify your answer.)
- **B.** There are infinitely many solutions and the equations are dependent. The solution set is $\{(x,y,z) | \square\}$.
- **C.** The system is inconsistent. The solution set is the empty set.

*6. Solve the system.

$$\begin{aligned} -9x - 3y + 4z - 4 &= 0 \\ 6x + 7y + 2z - 3 &= 0 \\ -3x - 5y + 6z - 16 &= 0 \end{aligned}$$

The solution set is $\{(\square,\square,\square)\}$.
(Type an integer or a simplified fraction.)

7. Jake has $184,000 to invest. He chooses one money market fund that pays 6.2% and a mutual fund that has more risk but has averaged 16.6% per year. If his goal is to average 8.8% per year with minimal risk, how much should he invest in each fund?

Jake should invest $☐ in the money market fund that pays 6.2%.
(Simplify your answer.)

Jake should invest $☐ in the mutual fund that averaged 16.6% per year.
(Simplify your answer.)

8. A glass of skim milk supplies 0.1 mg of iron and 8.5 g of protein. A quarter pound of lean meat provides 3.2 mg of iron and 23 g of protein. A person on a special diet is to have 3.6 mg of iron and 57.0 g of protein. How many glasses of skim milk and how many quarter-pound servings of meat will provide this?

How many glass(es) of skim milk and how many quarter-pound serving(s) of lean meat would be required?

☐ glass(es) of skim milk and ☐ quarter-pound serving(s) of lean meat will provide the needed nutrients.

9. A theater owner wants to divide a 3600 seat theater into three sections, with tickets costing $20, $50, and $100, depending on the section. He wants to have twice as many $20 tickets as the sum of the other tickets, and he wants to earn $133,000 from a full house. Find how many seats he should have in each section.

The number of $20 seats is ☐.

The number of $50 seats is ☐.

The number of $100 seats is ☐.

10. The nutritional content per ounce of three foods is presented in the table below. If a meal consisting of the three foods allows exactly 1000 calories, 51 grams of protein, and 3000 milligrams of vitamin C, how many ounces of each kind of food should be used?

	Calories	Protein (in grams)	Vitamin C (in milligrams)
Food A	100	13	350
Food B	200	9	750
Food C	300	11	400

Food A: ______ oz

Food B: ______ oz

Food C: ______ oz